

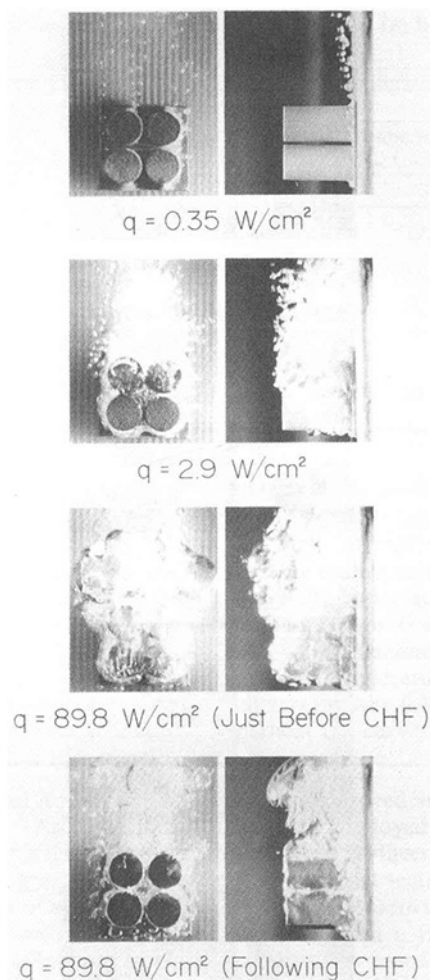


Fig. 8 Development of boiling pattern on the 4-pin surface in saturated FC-87 with increasing heat flux

lowing a periodic release of vapor in discrete volumes. The formation of a thin continuous vapor blanket on the perimeter and tip regions of the pins signaled transition to film boiling.

**Saturated Boiling on Pin-Fin Surfaces.** Figure 9 shows FC-87 boiling data for the long and short smooth studs displayed in Fig. 6. The slope of the natural convection line is greater than the predicted slope for a horizontal cylinder. End effects caused by the presence of the heat source module at the stud base may account for this discrepancy. In the nucleate boiling region the numerical solution fits the experimental data closely. This result lends credibility to the modeling assumption that localized boiling heat transfer from a non-isothermal surface is similar to that from an isothermal surface at the same temperature regardless of surface orientation. However, it should be emphasized that the numerical study was not specifically aimed at predicting CHF since the hydrodynamic conditions associated with CHF on a cylinder differ markedly from those on a vertical flat surface. Although the upper heat flux limit for the numerically-determined boiling curve corresponds to local enhanced surface conditions identical to those of a flat surface at CHF, the upper limit should merely be treated as signaling a tendency of the surface to promote local vapor blanketing.

Although the long stud had over twice the surface area of the short stud, Fig. 9 shows CHF for the long stud was only 30 percent higher. Based on photographic analysis of the boiling stud, it can be concluded that the modest enhancement is a result of local dryout near the base of the fin. As the base heat flux and superheat rose to a point great enough to sustain

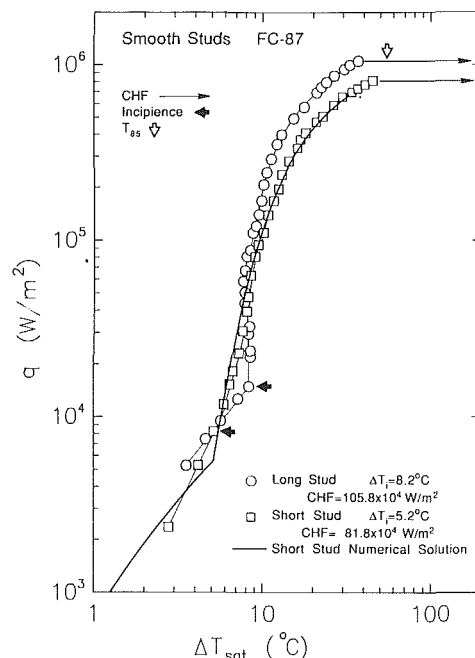


Fig. 9 Boiling curves for the long and short smooth studs in saturated FC-87

a small but stable vapor blanket, the insulating effect of the blanket introduced additional conduction resistance between the fin base and the now decreased nucleate boiling heat transfer area. Thus, the base superheat increased in response to the combined effects of increased internal conduction resistance and decreased heat transfer area. Further increase in base heat flux pushed the fin heat transfer equilibrium toward a condition where the vapor blanket area increased unrestrained and caused CHF.

The validity of superposition of cylindrical fin elements was tested using the 4 and 9-pin surfaces shown in Fig. 6. Experimental data in the nucleate boiling region for the 4-pin surface, Fig. 10(a), are closely predicted by the numerical model based on superposition of four identical elements, each carrying a smooth stud. The 9-pin surface, Fig. 10(b), apparently experienced more pin-to-pin interference than the 4-pin array since the 9-pin data is shifted by approximately  $2.5^{\circ}\text{C}$  to the left of the predicted curve. Nevertheless, the agreement is still acceptable for electronic packaging purposes.

**4.2 Multi-Level Surface Enhancement.** The surface enhancement results of the pin-fin surfaces reported in this paper as well as those of the low-profile microgroove and microstud surfaces and of other flat, textured surfaces (Anderson and Mudawar, 1989) all point to the potential for significant enhancement of boiling performance which can result from combining the various augmentation geometries into a single chip surface attachment. Figure 11 shows three surface attachments whose geometries were based upon three levels of enhancement: (1) a single extended cylindrical stud, (2) low profile microstructures machined into the stud perimeter, and (3) microscopic surface characteristics achieved by blasting the surface with an air/water/silica slurry prior to testing. The axial and radial microgroove enhanced studs were constructed by grooving a smooth stud along the axial and radial directions, respectively. The microstud-enhanced stud consisted of a microstud geometry machined into the smooth stud. The surface microstructures shown in Fig. 11 are identical to those of the low-profile reference surfaces (see Fig. 3). Figure 11 also shows various length scales associated with the microstructured stud attachments. The microscopic characteristics resulting from vapor blasting influence the incipience and low nucleation den-